



Pipe Flow Analysis

Bernoulli EQ

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g z_2$$

along a streamline
if the flow is steady
incompressible
without internal-energy change

Energy EQ (Generalized B.E.)

1st law of thermodynamics

$$\begin{aligned} \rightarrow \overset{\text{HT}}{q} &= \dot{m} \left[\overset{\text{enthalpy}}{(h_2 - h_1)} + \overset{\text{K.E.}}{\frac{1}{2} (V_2^2 - V_1^2)} + \overset{\text{P.E.}}{g(z_2 - z_1)} \right] + \overset{\text{shaft work}}{\dot{W}_s} \\ &= \dot{m} \left[\underset{\text{internal energy}}{(u_2 - u_1)} + \underset{\substack{\text{pressure (flow) work}}}{\frac{1}{\rho} (P_2 - P_1)}} + \overset{\text{K.E.}}{\frac{1}{2} (V_2^2 - V_1^2)} + \underset{\text{P.E.}}{g(z_2 - z_1)} \right] + \dot{W}_s \end{aligned}$$

$$\therefore \boxed{P_1 + \frac{1}{2} \rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g z_2 + w_L}$$

$$w_L \text{ (loss) } \equiv \rho(u_2 - u_1) - \frac{\rho q}{\dot{m}} = \rho(u_2 - u_1) - \frac{q}{Q}$$

or "pressure loss"

(NOTE) loss is due to heat transfer

internal energy change (friction)

Alternative form

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + \underset{\substack{\uparrow \\ \text{"head loss"}}}{h_L}$$



For a pipe flow with a non-uniform velocity distribution

$$\frac{P_1}{\rho g} + \frac{\alpha_1 \bar{V}_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{\alpha_2 \bar{V}_2^2}{2g} + z_2 + h_L$$

(α_1, α_2 : kinetic energy coefficient)

If. $\left. \begin{array}{l} \text{F.D. flow} \\ D_1 = D_2 \\ z_1 = z_2 \\ (V_1 = V_2) \end{array} \right\} \rightarrow \frac{\Delta P}{\rho g} = h_L \quad \Delta P = f \cdot \frac{1}{2} \rho V^2 \cdot \frac{L}{D}$

$$\therefore h_L = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

(Head loss by friction)

Major loss

Minor losses by system components

- sharp entrance / exit
- valve
- sudden expansion
- elbows
- tees
- ⋮



NOTE

Major loss is
NOT major
in magnitude!

Head loss due to minor losses

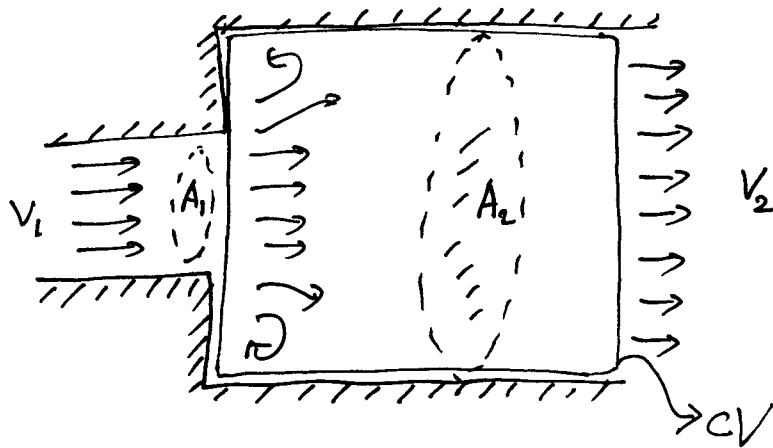
$$h_L = K_L \frac{V^2}{2g}$$

K_L : loss coefficient

(of $\underbrace{f \cdot \frac{L}{D}}_{\text{major loss}}$)

= f_m (geometry, Re)

Example loss coefficient for a sudden expansion



$$A_1 V_1 = A_2 V_2$$

$$P_1 A_2 - P_2 A_2 = \rho A_2 V_2 (V_2 - V_1)$$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + \underbrace{\left(\frac{h_L}{1} \right)}_{K_L \frac{V_1^2}{2g}}$$

$$\Rightarrow K_L = \left(1 - \frac{A_1}{A_2} \right)^2$$